



Tag 13
08.07.2026
mittwoch

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo <https://github.com/www42/a5>

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

Heute



What happened yesterday



Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services

Signing method ⓘ
☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1 ▼

Permissions * ⓘ

Read ▼

Start and expiry date/time ⓘ
Start

02/01/2021 

(UTC-08:00) Coordinated Universal Time-08 ▼

Expiry

02/02/2021 

(UTC-08:00) Coordinated Universal Time-08 ▼

Allowed IP addresses ⓘ

for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ
☒ HTTPS ☐ HTTP

[Generate SAS token and URL](#)

Mini-Übung

List and Rotate Storage Access Keys (Azure CLI)

Azure for Student

 GitHub a5

In this challenge, you will configure network peering between two Azure virtual networks. First, you will review the existing environment. Next, you will configure Azure virtual network peering. Finally, you will create two virtual machines by using Azure PowerShell, and then you will use the virtual machines to test the virtual network peering connections. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge...

Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

3



[Implement an Azure Load Balancer \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.2-007

⌚ 30 minutes

In this Challenge Lab, you will implement an Azure load balancer. First, you will create an Azure load balancer. Next, you will configure a backend pool for the load balancer, and then you will configure an HTTP health probe. Finally, you will create a load balancing rule. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge before you start.

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

8

Challenge Labs supporting AZ-104: Implement and manage storage in Azure

1



[Manage Access to Azure Storage \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.1-002

⌚ 30 minutes

In this Challenge Lab, you will secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using...

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

9

Challenge Labs supporting AZ-104: Deploy and manage Azure compute resources

1



[Enable Azure Virtual Machine Scale Sets for High Availability and Scalability \[Guided\]](#)

Challenge Labs All Access Pass , HCS-009

⌚ 30 minutes

In this challenge, you will create and deploy multiple Azure virtual machine scale sets acting as web server and app server tiers. First, you will create a virtual machine scale set for the web server tier. Next, you will use an Azure Resource Manager (ARM) template to create a virtual machine scale set for the app server tier. Finally, you will verify connectivity to the virtual machine scale set for the web server tier. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set...

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

10

Challenge Labs supporting AZ-104: Monitor and back up Azure resources

1



[Implement Azure Backup for Azure Virtual Machines \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.2-009

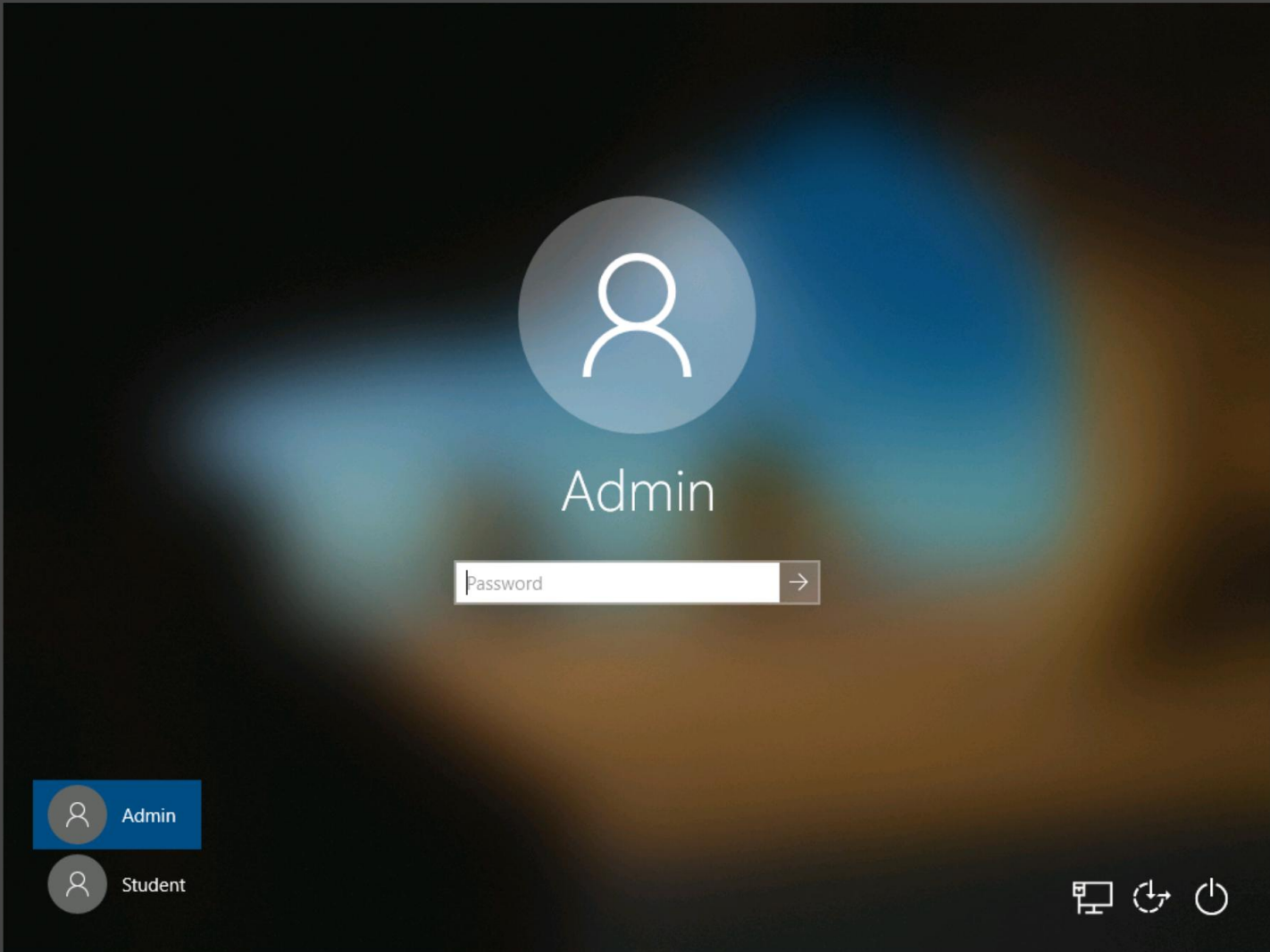
⌚ 30 minutes

In this challenge, you will implement Azure backup for your Azure virtual machines. First, you will back up an Azure virtual machine by using the Azure portal. Next, you will enable

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes

The login screen features a dark background with a blurred image of a person. In the center, there is a large, semi-transparent circular icon containing a white silhouette of a person. Below this icon, the word "Admin" is written in a large, white, sans-serif font. Underneath the name, there is a white rectangular input field with the placeholder text "Password" and a small grey arrow pointing to the right. In the bottom-left corner, there are two circular icons: the top one is blue with a white person silhouette and the label "Admin" next to it; the bottom one is grey with a white person silhouette and the label "Student" next to it. In the bottom-right corner, there are three small white icons: a monitor, a circular arrow, and a power button.

AZ300.1-002: Manage Access to Azure Storage [Guided]
30 Minutes Remaining

Instructions Resources Help

The logo for Skillable Challenges, featuring a green icon of a graduation cap with an upward-pointing arrow and the text "skillable challenges" in a bold, sans-serif font.

Guided

Manage Access to Azure Storage

Challenge Overview

Understand the scenario

You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.

Navigating the Challenge Lab

 Understanding a Cloud Slice 

 Quick tips for navigating the Challenge Lab instructions. 

Next: Configure a storage account to... >

The screenshot shows the Windows 10 desktop with the Settings app open to the 'Language' section. The left sidebar shows 'Time & Language' selected. The main pane shows 'Deutsch (Deutschland)' as the default app language, with 'English (United States)' as the Windows display language. A red circle highlights the 'Deutsch (Deutschland)' entry, and a red arrow points from the taskbar to the bottom right of the settings window.

AZ300.1-002: Manage Access to Azure Storage [Guided]
23 Minutes Remaining

Instructions Resources Help

100%

skilable challenges

Guided

Manage Access to Azure Storage

Challenge Overview

Understand the scenario

You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.

Navigating the Challenge Lab

- Understanding a Cloud Slice
- Quick tips for navigating the Challenge Lab instructions.

Next: Configure a storage account to...

AZ-104

 Deploy and manage
Azure compute resources



Introduction to Azure Virtual Machines



Review Cloud Services Responsibilities

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Microsoft	Shared	Customer	Customer
	Network controls	Microsoft	Shared	Customer	Customer
	Operating system	Microsoft	Microsoft	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

 Microsoft  Customer  Shared

Plan Virtual Machines

- Start with the network
- Name the virtual machine
- Choose a location
 - Each region has different hardware and service capabilities
 - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing



Determine Virtual Machine Sizing

Type	Description
General purpose	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

Determine Virtual Machine Storage

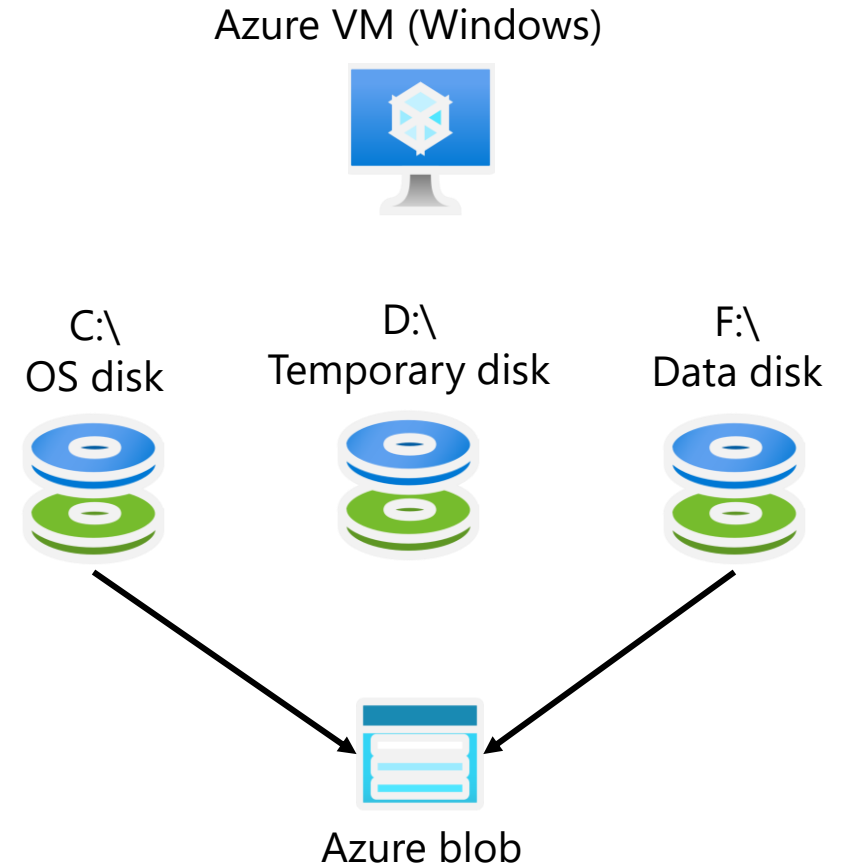
Each Azure VM has two or more disks:

- OS disk
- Temporary disk (not all SKUs have one, content can be lost)
- Data disks (optional)

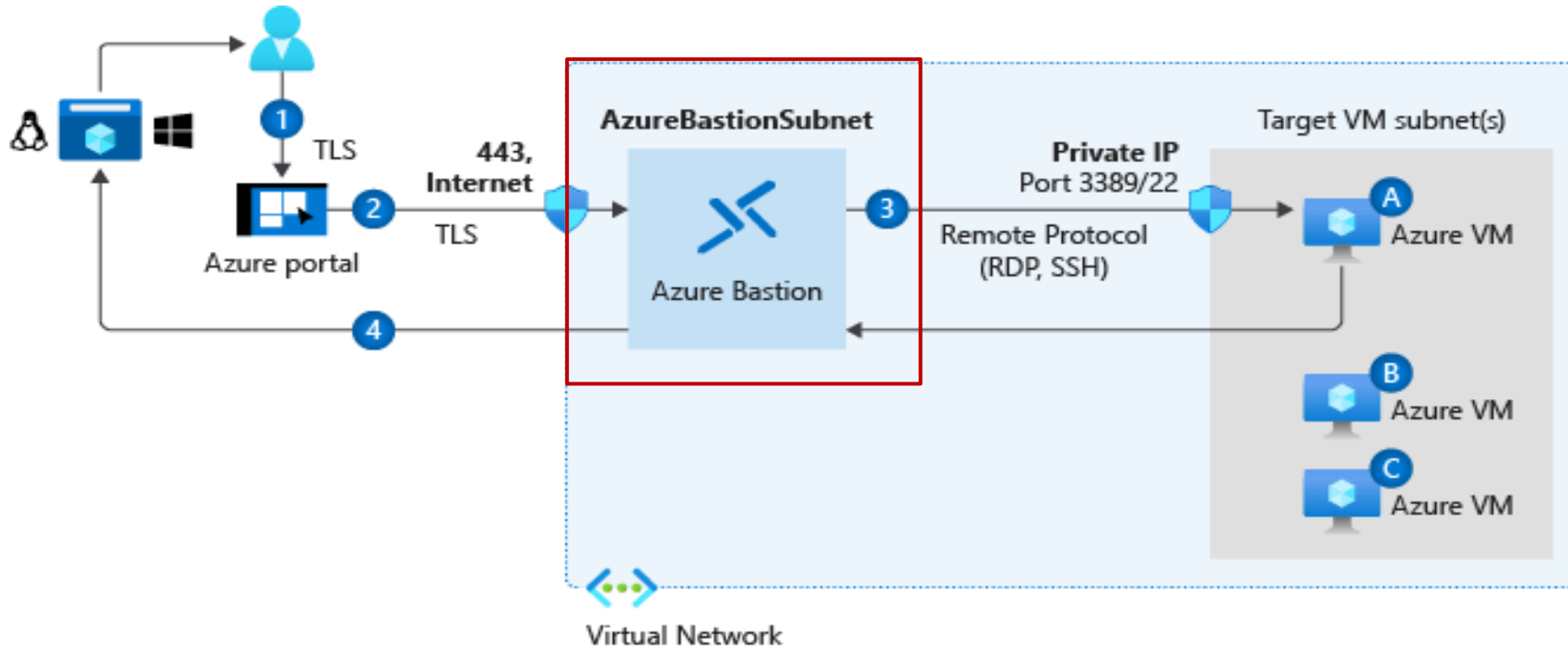
OS and data disks reside in Azure Storage accounts:

- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks



Connect to Virtual Machines



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Labs



TO THE LAB ZONE →

NO THROUGH ROUTE
Berlin
Europa
Dublin

SPEEDY ROUTE

4²

lounge
Microsoft
tech-ed
2010

lounge

software diagnostics

E 94

SD Studio
for Team Foundation Server
Visualize your code in
a software map.

SD Developer Edit
for C and C++ for Visual
Trace dynamically your code
Visualize the control flow.

software
diagnostics

DataCore
SOFTWARE

End of presentation



